Information Technology Department

Sample Final Examination

CSCN1104 – Computer Networking Fundamentals

Max Marks	40	Time Allocation	2 Hours

Student ID		Student Name	
Section		Level	Diploma 1st year

Instructions to the Students

1. Answer the questions in the space provided by using **blue or black pen** only. You will obtain a zero (0) if you use a pencil for all questions.
2. Use of mobile phones and any electronic gadgets inside the examination hall are not allowed.
3. ***Copying, Cheating, and any kind of malpractices in the examination are strictly prohibited.*** Action will be taken as per the "Article 80, Bylaws of Colleges of Technology".
4. Students should bring their own pencils, pens, erasers, rulers, calculators, and any other tools required for the exam.
5. Use of standard calculators is allowed for this exam.
6. The exam has a total of **32** non optional questions in all.
7. Exam paper contains **9** pages including the cover page.
8. Use the backside of the page for any required calculations.

Section	Allotted Marks	Obtained Marks
Section A	12.00	
Section B	12.00	
Section C	6.00	
Section D	10.00	
Total Marks	40.00	

Marked By	Checked/Re-marked by

SECTION-A: MULTIPLE CHOICE QUESTIONS (12 X 1 Mark = 12 Marks)

Select the best answer

#	Questions
1	**Which of the following is a type of network administered by one or more service providers?** a. Wireless Local Area Network b. Wide Area Network c. Local Area Network d. Personal Area Network
2	**In which of the following memory of a switch is the startup configuration stored?** a. NVRAM b. FLASH c. RAM d. All of the above
3	**Which of the following best describes the term "reliability" in a network?** a. Determines when someone can send a message b. Provides guaranteed delivery c. Determines if data is corrupted d. Ordering the data in proper order
4	**Which of the following type of copper cable provides better noise protection?** a. Shielded Twisted Pair (STP) b. Unshielded Twisted Pair (UTP) c. Both options a and b d. None of the options are correct
5	**Which of the following is an example of LAN topology?** a. Point-to-Point b. Star c. Hub and Spoke d. Mesh

6	**Which of the following address is used by a layer-2 switch to make its forwarding decision?** a. IPv4 address b. IPv6 address c. MAC address d. Port address
7	**Which of the following fields is a 32 bit binary value that represents the senders address in a packet header?** a. Source IPv4 address b. Source IPv6 address c. Destination IPv4 address d. Destination IPv6 address
8	**Which of the following is true on the use of the command "service password-encryption"?** a. It is used to encrypt only the currently available passwords on a device b. It is used to encrypt only the future configured passwords on a device c. It is used to encrypt both the currently available and future configured password on a device d. It is used to encrypt the service password

9	**Which of the following commands would generate the output shown below?**
	
	`Interface  IP-Address  OK? Method Status  Protocol`
	`GigabitEthernet0/0/0  192.168.10.1  YES manual up  up`
	`GigabitEthernet0/0/1  209.165.200.225 YES manual up  up`
	`Vlan1  unassigned  YES unset  administratively down down`
	
	a. Show ip route
	b. Show ipv4 route
	c. Show ip interface brief
	d. Show ipv4 interface brief

10	**Which of the following best describes the address 192.168.10.79/28?** a. It is a last usable host address in the subnet 192.168.10.80/28 b. It is the broadcast address of the subnet 192.168.10.64/28 c. It is a first usable host address in the subnet 192.168.10.78/28 d. It is a last usable host address in the subnet 192.168.10.64

11	**Which of the following type of information is included in the transport layer header?** a. Destination and source logical address b. Destination and source port numbers c. Destination and source physical address d. Destination and source PC numbers
12	**Which of the following is used to map the domain names to its respective IP addresses?** a. DHCP b. FTP c. HTTP d. DNS

SECTION-B: FILL IN THE BLANKS (12 X 1 Mark = 12 Marks)

#	Questions
13	___Internet_____ is a worldwide collection of interconnected LANs and WANs.
14	___Telnet/SSH_____is a method used to establish a remote CLI connection to a device.
15	The process of transmitting a message from one source to one destination device is referred as ____Unicast_____.
16	_____RJ-45_____is a type of connector used to terminate STP cable.
17	The _____MAC_____sub layer of data link layer is responsible for media access control.
18	The second part of the MAC address is referred as _____ vendor-assigned _____.
19	The command that is used to view the routing table of a windows PC is ___show ip route____.
20	The command that is used to encrypt both the currently available and future configured password on a router is ___service password-encryption_____.

21	The number of usable host addresses, if one bit is borrowed from the host portion of a class-C address is _____126_____.
22	In an IPv6 address, a contiguous string of zeros can be written as __::_____.
23	The combination of source IP address and Source Port number is called a __socket_____.
24	The physical and data link layer of OSI model is equivalent to the __Network Access_____ layer of TCP/IP model.

Section–C: VERY SHORT ANSWER QUESTIONS (6 Marks)

25. Write down the commands on the router to accomplish the tasks given.

#	Question	Answer	Marks
1	Configure the device name on a switch as H12LAB	Hostname H12LAB	0.25
2	Protect the access to the user mode using the password ER@#TYab	Line console 0 Password ER@#TYab login	0.75
3	Protect the remote access to the device using the password KG#@12ds	Line vty 0 15 Password KG#@12ds Login	0.75
4	Assign the last usable IP address in the subnet 192.168.15.96/29 to VLAN 1 interface of switch and enable it	Interface VLAN 1 ip address 192.168.15.102 255.255.255.248 No shutdown	0.75
5	Assign the fifth usable address in the subnet 192.168.15.96/29 as the gateway for the switch	ip default-gateway 192.168.15.101	0.50

26. By using the appropriate subnetting method, subnet the address 192.168.15.0/24 for the required number of usable host addresses as given. LAB 1 network requires 19 usable host addresses, LAB 2 network requires 9 usable host addresses and ISP network requires 2 usable host addresses. Use the following instructions when using the subnets and complete the subnet table given below. (3)

i. Use the second available subnet to the network with the highest host requirement
ii. Use the next available subnet to the network with the second highest host requirement
iii. Use the next available subnet to the network with the third highest host requirement

Subnet Table:

Network Name	Network Address	First Usable Host Address	Second Usable Host Address	Last Usable Host Address	Broadcast Address	Subnet Mask
LAB 1	192.168.15.0	15.1	15.2	15.30	15.31	.224
LAB 2	192.168.15.32	15.33	15.34	15.46	15.47	.240
ISP	192.168.15.48	15.49	NA	15.50	15.51	.252

a	What is the second usable host address in LAB 1 subnet?	192.168.15.2
b	What is the last usable host address in LAB 1 subnet?	192.168.15.30
c	What is the subnet mask for LAB 2 subnet?	255.255.255.240
d	What is the first usable host address in LAB 2 subnet?	192.168.15.33
e	What is the network address for ISP subnet?	192.168.15.48
f	What is the broadcast address for ISP subnet?	192.168.15.51

SECTION-D: SHORT ANSWER QUESTIONS (10 Marks)

27. Let's suppose there is an organization with two departments with each department requiring 100 host addresses. The administrator has used 192.168.10.0/24 and 192.168.11.0/24 networks. Briefly explain in terms of usage of addresses what is the problem of using two separate networks and how this issue can be resolved? (2)

Each class-C network has a total of 254 usable addresses available. By using only 100 address we are wasting 154 addresses in each network. A total of 308 addresses are wasted from both the networks. Instead of using two separate networks, we can use only one network and perform subnetting which will help in reducing the wastage of address.

28. Briefly explain at least one function for each of the following layers of OSI model along with the name of the PDU at each layer. (2)

Layer	Function	PDU
Presentation	Provides for common representation of the data transferred between application layer services.	Data
Transport	Defines services to segment, transfer, and reassemble the data for individual communications.	Segment
Network	Provides services to exchange the individual pieces of data over the network.	Packet
Physical	Describes the means to activate, maintain, and de-activate physical connections.	Bits

29. Differentiate between the physical and logical topology. (1)

Physical topology diagrams illustrate the physical location of intermediary devices and cable installation.

Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network.

30. Explain the use of the following HTTP message types. (1)

GET: This is a client request for data. A client (web browser) sends the GET message to the web server to request HTML pages.

POST: This uploads data files to the web server, such as form data.

31. Write down at least four differences between TCP and UDP. (2)

TCP	UDP
TCP is a connection-oriented protocol	UDP is a connectionless protocol.
Ensures Reliable Delivery	Does not ensure reliability of data
Lost segments are retransmitted	Any segments that are lost are not resent.
Provides Same-Order Delivery	Data is reconstructed in the order that it is received.

32. Refer to the topology given. There are two PCs. PC1 is given an address of 192.168.75.126 and PC2 is given an address of 192.168.75.129. The value for the subnet mask for both the PCs is /29. Answer the following questions. (2)

Question #	Question	Answer
1	What is the network address for PC1?	192.168.75.120
2	What is the network address for PC2?	192.168.75.128
3	Will these two PCs be able to communicate with each other directly without the use of a router?	No
4	How can you make these two PCs communicate directly with each other without the use of a router?	By getting both the PCs in the same network. Both the PCs can be in 192.168.75.120 or 192.168.75.128 network